DALE ZHOU

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Education

University of Pennsylvania — 2023
Ph.D in Neuroscience
Advisors: Dani Bassett & Theodore Satterthwaite

University of Maryland, College Park — 2015
B.A. (honors) in Philosophy
B.S. (honors) in Psychology
Minor in Neuroscience

Academic Appointments

Hewitt Postdoctoral Research Fellow — 2023–Present
University of California, Irvine
Advsiors: Aaron Bornstein & Michael Yassa

Postbaccalaureate Researcher (IRTA) — 2017
National Institute of Mental Health
Child Psychiatry Branch (PI: Judy Rapoport)

Interests

Network science, computational psychiatry, control theory, memory, cognitive control, curiosity, foraging, dynamical systems, representations, information theory, reinforcement learning

FUNDING

Active

2027–2029 Brain & Behavior Research Foundation NARSAD Young Investigator Grant — $70,000
Reward Timing Adaptations to Early Life Unpredictability in Anhedonia.

2023–2026 Hewitt Foundation Fellowship — $261,000
Information Compression in Reward and Memory Dysfunction.
Salary and stipend from the George E. Hewitt Foundation for Medical Research to support the next generation of biomedical research leaders at UC Irvine, Salk Institute, & Scripps Research Institute.

Completed

2021–2023 NIH Predoctoral National Research Service Award — $80,138
Brain Network Maturation and Executive Dysfunction Spanning Diagnostic Categories of Psychopathology.
Grant # F31MH126569

AWARDS & HONORS

2026	Neuroscience and Philosophy Seminar Honorarium	Duke
2024	Summer Institute in Neuroscience Mentor Stipend	UC Irvine
2023	Biomedical Graduate Studies Course Funds	UPenn
2022	Biomedical Graduate Studies Travel Funds	UPenn
2019	National Academy of Sciences Travel Award	NAS
2018–20	Language and Communication Sciences Stipend	UPenn
2015–17	NIMH Intramural Research and Training Award	NIH
2015	Honors in Psychology	UMaryland
2015	Honors in Philosophy	UMaryland
2013	College Park Scholars Co-Curricular Scholarship	UMaryland
2011	Ling Ho Anita K'ung Tong Scholarship	UMaryland
2010–14	President's Scholarship (4-year merit scholarship)	UMaryland
2010–12	Scholar in Global Public Health	UMaryland

PUBLICATIONS

Google Scholar: citations = 1912 | h-index = 16 | i10-index = 22

Submitted/ Under Revision

* = mentee
† = co-first
‡ = co-last

[6] Mohammed, N., **Zhou, D.**, Bassett, D.S., Zurn, P., Lingel, J., & Lydon-Staley, D.M. (2026). Busybody, Hunter, and Dancer: Curious (social) practices in creativity. *PsyRxiv*

[5] Irizarry-Martínez, G.[*], Adams, J.N. Leonard, BT[*], **Zhou, D.**, Bock, J., Granger, S.J., McMillan, L, Yassa, M. (2026). Functional connectivity of hippocampal-subcortical circuits differ by sex in major depressive disorder.

[4] **Zhou, D.**, Noh, S., Harhen, N., Banavar, N., Kirwan, B., Yassa, M.A, & Bornstein, A.M. (2025). A compressed code for memory discrimination. *bioRxiv*: 10.12.681901

[3] **Zhou, D.**, Cosme, D., Kang, Y., Stanoi, O., Lydon-Staley, D.M., Mucha, P., Falk, E., Ochsner, K., & Bassett, D.S (2025). Neural dynamics of cognitive control: Current tensions and future promise. *PsyArXiv*: 10.31234/osf.io/ za7yx

[2] **Zhou, D.**, Ahn, J., Lydon-Staley, D.M., Falk, E.B., Bassett, D.S.[‡], & Ruscio, A.M.[‡](2025). Neural dynamics underlying the persistence of perseverative thought in depression and anxiety. *bioRxiv*: 10.1101/2025.07. 21.666019

[1] Adams, J.N., Kark, S.M., Chappel-Farley, M.G., Escalante, Y., **Zhou, D.**, Stith, L.A., Rapp, P.E., Yassa, M.A., & the Alzheimer's Disease Neuroimaging Initiative (2024). Pathological brain state dynamics in Alzheimer's disease. *bioRxiv*: 10.1101/2023.08.30.555617

Journal Articles

[18] Noh, S.M.[†], **Zhou, D.**[†], Cooper, K., Guo, S.[*], Dinh, E.[*], Bornstein, A.M. (2026). Sparse and distributed memory constraints modulate the representational change of learning strategies. *Neuropsychologia*

↪ † = co-first author

[17] Noh, S., Cooper, K., Guo, S.[*], **Zhou, D.**, Stark, C., Bornstein, A.M. (2025). *Journal of Gerontology: Psychological Sciences*. Multi-step planning across the human lifespan can be improved with individualized memory interventions. *The Journals of Gerontology: Series B*. DOI:10.1093/geronb/gbag038

[16] **Zhou, D.**, Patankar, S.P., Lydon-Staley, D.M., Zurn, P., Gerlach, M.[‡], & Bassett, D.S.[‡] (2024). Architectural styles of curiosity in global Wikipedia mobile app readership. *Science Advances*. DOI: 10.1126/sciadv.adn3268

↪ Featured in *Nature* | *Science (podcast)* | *Scientific American*

[15] Parkes, L., Kim, J.Z., Stiso, J., Brynildsen, J.K., Cieslak, M., Covitz, S., Gur, R.E., Gur, R.C., Pasqualetti, F. Shinohara, R.T., Stiso, J., **Zhou, D.**, Satterthwaite, T.D., & Bassett, D.S. (2023). Using network control theory to study the dynamics of the structural connectome. *Nature Protocols*. DOI: 10.1038/s41596-024-01023-w

[14] Kang, Y., Ahn, J., Cosme, D., McGowan, A., Mwilambwe-tshilobo, L., **Zhou, D.**, Jovanova, M., Stanoi, O., Mucha, P.J., Ochsner, K.N., Bassett, D.S., Lydon-Staley, D. & Falk, E.B. (2023). Frontoparietal functional connectivity moderates the link between time spent on social media and subsequent negative affect in daily life. *Scientific Reports*. DOI: 10.1038/s41598-023-46040-z

[13] Mahadevan, A., Cornblath, E., Lydon-Staley, D.M., **Zhou, D.**, Parkes, L., Larsen, B., Adebimpe, A., Kahn, A.E., Gur, R.C., Gur, R.E., Satterthwaite, T.D., Wolf, D.H., & Bassett, D.S. (2023). Alprazolam modulates persistence energy during emotion processing in first-degree relatives of individuals with schizophrenia: a network control study. *Molecular Psychiatry*. DOI: 10.1038/s41380-023-02121-z

[12] **Zhou, D.**, Kang, Y., Cosme, D. Jovanova, M., He, X., Mahadevan, A., Ahn, J., Stanoi, O., Brynildsen, J.K., Cooper, N., Cornblath, E.J., Parkes, L., Mucha, P., Ochsner, K., Lydon-Staley, D., Falk, E., & Bassett, D.S. (2023). Mindful Attention Promotes Control of Brain Network Dynamics for Self-Regulation and Discontinues the Past from the Present. *PNAS*. DOI: 10.1073/pnas.220107411

[11] Patankar, S.P., **Zhou, D.**, Lynn, C.W., Kim, J., Ouellet, M., Ju, H., Zurn, P., Lydon-Staley, D.M., & Bassett, D.S. (2022). Curiosity as filling, compressing, and reconfiguring knowledge networks. *Collective Intelligence*. *arXiv*: 10.48550/arXiv.2204.01182

[10] Richie-Halford, A., Cieslak, M., Ai, L., Caffarra, S., Covitz, S., Franco, A.R., Karipidis, I.I., Kruper, J., Milham, M., Avelar-Pereira, B., Roy, E., Sydnor, V.J., Yeatman, J.D., [The Fibr Community Science Consortium, including **Zhou, D.**], Satterthwaite, T.D., and Rokem, A. (2022). An open, analysis-ready, and quality controlled resource for pediatric brain white-matter research. *Scientific Data*. DOI: 10.1038/s41597-022-01695-7

[9] Ju, H., **Zhou, D.**, Blevins, A.S., Lydon-Staley, D.M., Kaplan, J., Tuma, J.R., Bassett, D.S. (2022). Historical growth of concept networks in Wikipedia. *Collective Intelligence*. DOI: 10.1177/26339137221109839

[8] Weninger, L., Srivastava, P., **Zhou, D.**, Kim, J.Z., Cornblath, E.J., Bertolero, M.A., Habel, U., Merhof, D., and Bassett, D.S (2022). The information content of brain states is explained by structural constraints on state energetics. *Physical Review E*. DOI: 10.1103/PhysRevE.106.014401

[7] Adebimpe, A., Bertolero, M.A, [33 others, including **Zhou, D.**, and the ALLFTD Consortium], & Satterthwaite, T.D. (2022). ASLPrep: A Platform for Processing of Arterial Spin Labeled MRI and Quantification of Regional Brain Perfusion. *Nature Methods*. DOI: 10.1038/s41592-022-01458-7

[6] **Zhou, D.**, Lynn, C.W., Cui, Z., Ciric, R., Baum, G.L., Moore, T.M., Roalf, D.R., Detre, J.A., Gur, R.C., Gur, R.E., Satterthwaite, T.D., & Bassett, D.S (2021). Efficient Coding in the Economics of Human Brain Connectomics. *Network Neuroscience.* DOI: 10.1162/netn_a_00223

[5] Wang, X., Dworkin, J.D., **Zhou, D.**, Stiso, J., Falk, E.B., Bassett, D.S., Zurn, P., Lydon-Staley, D.M. (2021). Gendered citation practices in the field of communication. *Annals of the International Communication Association*. DOI: 10.31234/osf.io/ywrcq

[4] Lydon-Staley, D.M., **Zhou, D.**, Blevins, A.S., Zurn, P., & Bassett, D.S. (2020). Hunters, busybodies, and the knowledge network building associated with deprivation curiosity. *Nature Human Behavior.* DOI: 10.1038/s41562-020-00985-7

[3] Chai, L.R., **Zhou, D.**, & Bassett, D.S. (2019). Evolution of semantic networks in biomedical texts. *Journal of Complex Networks.* DOI: 10.1093/comnet/cnz023

[2] **Zhou, D.**, Liu, S., Zhou, X., Berman, R.A., Broadnax, D.D., Rapoport, J.L, & Thomas, A.G. (2018) 7 Tesla MRI reveals hippocampal structural abnormalities associated with memory intrusions in childhood-onset schizophrenia. *Schizophrenia Research.* DOI: 10.1016/j.schres.2018.07.023

[1] **Zhou, D.**, Gochman, P., Broadnax, D.D., Rapoport, J.L., & Ahn, K. (2016). 15q13.3 duplication in two patients with childhood-onset schizophrenia. *American Journal of Medical Genetics Part B: Neuropsychiatric Genetics.* DOI: 10.1002/ajmg.b.32439

Reviews, Commentary, & Book Chapters

[7] **Zhou, D.** & Bornstein, A.M. (2026). Decompressing the kinds of compression (Commentary on Dubova & Sloman). *Behavioral and Brain Sciences*.

[6] **Zhou, D.** & Bornstein, A.M. (2024). Expanding horizons in reinforcement learning for curious exploration and creative planning. *Behavioral and Brain Sciences*. *PsyArXiv*: 10.31234/osf.io/bhcwp

[5] Zurn, P., **Zhou, D.**, Lydon-Staley, D.M., & Bassett, D.S. (2022). Edgework: Viewing curiosity as fundamentally relational. In Cogliati Dezza, I., Wu, C., & Schulz, E. (Eds.). *The Drive for Knowledge: The Science of Human Information Seeking.* Cambridge University Press. *PsyArXiv*: 10.31234/osf.io/crzae

[4] **Zhou, D.**, Lydon-Staley, D.M., Zurn, P., & Bassett, D.S. (2020). The growth and form of knowledge networks by kinesthetic curiosity. *Current Opinion in Behavioral Sciences.* DOI:10.1016/j.cobeha.2020.09.007

[3] Srivastava, P., Nozari, E., Kim, J.Z., Ju, H., **Zhou, D.**, Becker, C., Pasqualetti, F., & Bassett, D.S. (2020). Models of communication and control for brain networks: distinctions, convergence, and future outlook. *Network Neuroscience.* DOI: 10.1162/netn_a_00158

[2] Rapoport, J. L., **Zhou, D.**, & Ahn, K. (2020). Intellectual disabilities. *New Oxford Textbook of Psychiatry, 3rd edition.* Oxford University Press, USA. ISBN: 9780198713005

[1] **Zhou, D.**, Sequeira, S., Driver, D., & Thomas, S. (2018). Disruptive Mood Dysregulation Disorder. In S. Thomas and D. Driver (Eds.), *Complex Disorders in Pediatric Psychiatry: A Clinician's Guide.* Clinics Review Articles, Elsevier Inc. ISBN: 9780323511476

Refereed Conference Papers

[4] **Zhou, D.**, Irizarry-Martínez, G.[*], Guo, S.[*], Kim, S., Janecek, J., Tuteja, N., Meza, N., McMillan, L., Thayer, J., Bornstein, A.M., Yassa, M.A. Effects of surprising rewards on pattern separation. Proceedings of the Annual Meeting of the Cognitive Science Society.

[3] Kang, Y., Ahn, J., Cosme, D., McGowan, A., Mwilambwe-tshilobo, L., **Zhou, D.**, Jovanova, M., Stanoi, O., Mucha, P.J., Ochsner, K.N., Bassett, D.S., Lydon-Staley, D. & Falk, E.B. Frontoparietal system functional connectivity moderates the within-day associations between increases in time spent on social media and subsequent negative affect. *73rd Annual International Communication Association Conference.* Toronto, CA. May 25-29, 2023.

↪ Promising Paper Award

[2] **Zhou, D.**, Kim, J.Z., Pines, A., Sydnor, V.J., Roalf, D.R., Detre, J.A., Gur, R.C., Gur, R.E., Satterthwaite, T.D., & Bassett, D.S. Compression supports low-dimensional representations of behavior across neural circuits. *NeurIPS 2022 Workshop on Information-Theoretic Principles in Cognitive Systems*. New Orleans, LA. December 3, 2022.

↪ Selected for oral talk (< 10% submissions)

[1] Wang, X., Lydon-Staley, D.M., Stiso, J.A., **Zhou, D.**, Falk, E.B, Bassett, D.S, Zurn, P. Gendered citation practices in the field of communication. *71st Annual International Communication Association Conference.* (virtual due to COVID-19). May 27-31, 2021.

↪ Communication & Science Biology Top Paper Award

In Prep

[4] **Zhou, D.**, Harhen, N.C., Stout, D.M., Jirsaraie, R.J., Parker, D.M., Yoo, J.[*], Risbrough, V., Davis, E.P., Glynn, L.M., Baram, T.Z., Yassa, M.A., Bornstein, A.M. Adaptation to early life unpredictability: elongated temporal representations for reward-based decision making.

[3] **Zhou, D.**, Irizarry-Martínez, G.[*], Guo, S.[*], Kim, S., Janecek, J., Tuteja, N., Meza, N., McMillan, L., Thayer, J., Bornstein, A.M., Yassa, M.A. Effects of surprising rewards on pattern separation.

[2] **Zhou, D.**, Tseytlin, I.[*], Satterthwaite, T.D., & Bassett, D.S. Predictive coding from compression, control, and recurrent connectivity in human brain networks.

[1] **Zhou, D.**, Kim, J.Z., Pines, A., Sydnor, V.J., Roalf, D.R., Gur, R.C., Gur, R.E., Satterthwaite, T.D., & Bassett, D.S. Network hubs compress sensation to cognition.

PRESENTATIONS

Invited Talks

2026	*Curiosity and Compression.* Cognition Seminar, invited by Melisa Gumus	UC Berkeley
2025	*Curiosity and Compression as Complex Systems.* Seminar, invited by Marina Dubova	Santa Fe Institute
2025	*Architectural styles of curiosity in global Wikipedia mobile app readers.* Research Showcase, invited by Wikimedia Research Team	Virtual
2025	*A lossy compression account of pattern separation challenges in aging.* Cognitive Science Colloquium, invited by student co-ordinators (internal)	UC Irvine
2023	*Compression, cognitive control, and curiosity.* Yassa Lab; Bornstein Lab (postdoc job talk)	UC Irvine
2023	*Compression, cognitive control, and curiosity.* Poldrack Lab (postdoc job talk; virtual)	Stanford

Submitted Talks & Symposia

2026	*Adaptation to early life unpredictability: elongated temporal representations for reward-based decision making. In symposium: Computational Mechanisms of Motivated Behavior across Development.* Flux Society	San Diego
2026	*Representation learning and choice in naturalistically complex environments.* West Coast Neuroeconomics Symposium	UCLA
2026	*A compressed code for memory discrimination.* Context & Episodic Memory Symposium	UPenn
2026	*Early-Life Unpredictability and the Neural and Computational Timescales of Integration. In symposium: From Unpredictability to Psychopathology: How Childhood Adversity Calibrates Reward Circuits and Risk for Mood and Anxiety Disorders.* Anxiety & Depression Association of America	Chicago
2025	*A lossy compression account of mnemonic discrimination: Computational theory and multimodal evidence.* Bay Area Memory Meeting	UC Davis
2024	*Architectural styles of curiosity in global Wikipedia mobile app readership.* 11th Annual Wiki Workshop - Research Track (virtual)	Virtual
2022	*Compression supports low-dimensional representations of behavior across neural circuits.* NeurIPS Workshop on Information-Theoretic Principles in Cognitive Systems ↪ < 10% submissions selected	New Orleans

Guest Presentations

2025	*A compressed code for memory discrimination* FeldmanHall Lab meeting (virtual)	Brown
2025	*A lossy compression account of pattern separation.* Mattar Lab meeting (virtual)	NYU
2023	*Compression: relationships to the brain, behavior, and transdiagnostic symptoms.* Astle Lab meeting (virtual)	Cambridge

Lightning Talks

2026	Computational Psychiatry Poster Spotlight ↪ < 10% submissions selected	Yale
2024	Hewitt Foundation Fellows Symposium	Salk Institute
2023	International Research Training Group	RWTH Aachen
2019	Brain Produces the Mind By Modeling	NAS
2017	Ninth Annual Julius Axelrod Symposium	NIH

Posters

[32] **Zhou, D.**, Guo, S.[*] (presenting author), Yassa, M.A., & Bornstein, A.M. Representation learning and choice in naturalistically complex environments. Cognitive Computational Neuroscience. New York, New York. August 3–6, 2026.

[31] **Zhou, D.**, Harhen, N.C., Stout, D.M., Jirsaraie, R.J., Parker, D.M., Yoo, J.[*], Risbrough, V., Davis, E.P., Glynn, L.M., Baram, T.Z., Yassa, M.A., Bornstein, A.M. Adaptation to early life unpredictability: elongated temporal representations for reward-based decision making. Computational Psychiatry Conference. Yale University, New Haven, CT. July 14–16, 2026.

↪ Poster spotlight (< 10% selected)

[30] **Zhou, D.**, Irizarry-Martínez, G.[*], Guo, S.[*], Kim, S., Janecek, J., Tuteja, N., Meza, N., McMillan, L., Thayer, J., Bornstein, A.M., Yassa, M.A. Effects of surprising rewards on pattern separation. Cognitive Science Society. Rio de Janeiro, Brazil. July 22–25, 2026.

[29] Mohammed, N., **Zhou, D.**, Bassett, D.S., Zurn, P., Lingel, J., Lydon-Staley, D.M. A role for overt social curiosity in creativity. Society for Psychology of Aesthetics, Creativity, and the Arts (Division 10 of the American Psychological Association). Omaha, Nebraska. March 12–14, 2026.

[28] Yoo, J.[*], Goeschel, A., **Zhou, D.**, Bornstein, A.M. Contingency-dependent state augmentation as a normative learning rule for non-Markovian tasks. *Society for Neuroscience*. San Diego, California. November 15–19, 2025.

[27] **Zhou, D.**[†], Noh, S.M.[†], Cooper, K.W., Guo, S.[*], Dinh, E.D.[*], & Bornstein, A.M. Sparse distributed memory constraints drive representational change as a function of temporal learning sequence. Cognitive Science Society. San Francisco, California. July 30–August 2, 2025.

[26] **Zhou, D.**, Ahn, J., Lydon-Staley, D.M., Falk, E.B., Bassett, D.S.[‡], & Ruscio, A.M.[‡], Network control dynamics of persistent thought patterns in depression and anxiety. Computational Psychiatry Conference. Tübingen, Germany. July 14–16, 2025.

[25] **Zhou, D.**, Ahn, J., Lydon-Staley, D.M., Falk, E.B., Bassett, D.S.[‡], & Ruscio, A.M.[‡], Network control dynamics of persistent thought patterns in depression and anxiety. Conte Center at UCI 12th Annual Symposium. Irvine, California. February 18, 2025.

[24] Irizarry-Martínez, G.[*], Leonard, B.T.[*], Adams, J.N., **Zhou, D.**, Granger, S., McMillan, L., Yassa, M.A. Resting-State Connectivity between the Locus Coeruleus and the Hippocampus differs by Sex and Depression diagnosis. *American College of Neuropsychopharmacology*. Phoenix, Arizona. December 8–11, 2024.

[23] Leonard, B.T.[*], Rasmussen, J, **Zhou, D.**, Small, S.L., Sandman, C.A., Stern, H., Baram, T.Z., Glynn, L.M., Poggi-Davis, E., Yassa, M.A. Early life adversity and paraventricular nucleus of thalamus network connectivity interact in the neurobiology of adolescent mental health. *American College of Neuropsychopharmacology*. Phoenix, Arizona. December 8–11, 2024.

[22] Mohammed, N., **Zhou, D.**, Bassett, D.S., Zurn, P., Lingel, J., Lydon-Staley, D.M. Dancing with Curiosity: A Case Study on the Role of Curiosity in the Creative Process. Society for Psychology of Aesthetics, Creativity, and the Arts. New Haven, Connecticut. March 13–15, 2025.

[21] Dinh, E.[*,†], **Zhou, D.**[†], Guo, S., Noh, S.M., Cooper, K., Bornstein, A.M. Autoencoder models of human graph learning reveal that sparse and dense representations differentially support planning and recall. *Society for Neuroscience*. Chicago, Illinois. October 5–9, 2024.

[20] **Zhou, D.**, Noh, S.M., Yassa, M.A., Bornstein, A.M. Pattern separation using compressed and semantic representations of memory. *Conference on Cognitive Computational Neuroscience*. Boston, Massachusetts. August 6–9, 2024.

[19] Yoo, J.[*], **Zhou, D.**, Bornstein, A.M. Latent cause inference as an efficient and flexible learning rule for cognitive graphs. *Conference on Cognitive Computational Neuroscience*. Boston, Massachusetts. August 6–9, 2024.

[18] Irizarry-Martínez, G.[*], Leonard, B.T.[*], Adams, J.N., **Zhou, D.**, Granger, S., McMillan, L., Yassa, M.A. Associations between resting-state functional connectivity of the locus coeruleus and anhedonia symptoms in individuals with depression. *International Behavioral Neuroscience Society*. Panama City, Panama. June 11–16, 2024.

[17] Parkes, L., Kim, J.Z., Stiso, J., Brynildsen, J.K., Cieslak, M., Covitz, S., Gur, R.E., Gur, R.C., Pasqualetti, F. Shinohara, R.T., Stiso, J., **Zhou, D.**, Satterthwaite, T.D., & Bassett, D.S. (2023). Network control theory (NCT) for neuroscientists: a Python-based protocol. *Organization for Human Brain Mapping*. Seoul, South Korea. June 23–June 27, 2024.

[16] **Zhou, D.**, Tseytlin, I.[*], Satterthwaite, T.D., & Bassett, D.S. (2023). Predictive coding from compression, control, and recurrent connectivity in human brain networks. *Conference on Cognitive Computational Neuroscience*. Oxford University, Oxford, England. August 24–27, 2023.

[15] Gataviņš, M.[*], Luo, A., Sydnor, V.J., Shafiei, G., **Zhou, D.**, Gur, R.E., Gur, R.C., Mackey, A.P., Satterthwaite, T.D., Keller, A.S. (2023). Functional network development along the sensorimotor-association axis. *Flux Society*. Santa Rosa, California. September 6–9, 2023.

[14] **Zhou, D.**, Patankar, S., Gerlach, M., Lydon-Staley, D.M., Zurn, P., & Bassett, D.S. Dynamics Of Curiosity And Complexity In Wikipedia Readers. Curiosity, Creativity and Complexity conference (unable to attend). Columbia University, New York City, New York. May 23–25 2023.

[13] **Zhou, D.**, Kim, J.Z., Pines, A., Sydnor, V.J., Roalf, D.R., Detre, J.A., Gur, R.C., Gur, R.E., Satterthwaite, T.D., & Bassett, D.S. Compression supports low-dimensional representations of behavior across neural circuits. *NeurIPS 2022 Workshop on Information-Theoretic Principles in Cognitive Systems*. New Orleans, LA. December 3, 2022.

[12] Brynildsen, J. K., **Zhou, D.**, Cosme, D., Jovanova, M., He, X., Mucha, P.J., Ochsner, K.N., Lydon-Staley, D.M. , Falk, E. B., Bassett, D.S. Regulation of alcohol cue reactivity in a social context. *Society for Neuroscience*. San Diego, CA. November 12, 2022.

[11] **Zhou, D.**, Lynn, C.W., Cui, Z., Ciric, R., Baum, G.L., Moore, T.M., Roalf, D.R., Detre, J.A., Gur, R.C., Gur, R.E., Satterthwaite, T.D., & Bassett, D.S. Network Fidelity Improves with Brain Network Maturation and Executive Function. *Flux Society*. Paris, France. September 6–9, 2022.

[10] **Zhou, D.**, Kim, J.Z., Pines, A., Sydnor, V.J., Roalf, D.R., Detre, J.A., Gur, R.C., Gur, R.E., Satterthwaite, T.D., & Bassett, D.S. Communication and compression principles integrate sensation to cognition in human brain networks. *Organization for Human Brain Mapping*. Glasgow, Scotland. June 19–23, 2022.

[9] **Zhou, D.**, Kang, Y., Cosme, D. Jovanova, M., He, X., Mahadevan, A., Stanoi, O., Brynildsen, J.K., Cooper, N., Cornblath, E.J., Parkes, L., Mucha, P., Ochsner, K., Lydon-Staley, D., Falk, E., and Bassett, D.S. Mindfulness Promotes Control of Network Dynamics for Self-Regulation and Updates the Past to Present. *Organization for Human Brain Mapping*. Glasgow, Scotland. June 19–23, 2022.

[8] Ju, H., **Zhou, D.**, Blevins, A.S., Lydon-Staley, D.M., Kaplan, J., Tuma, J.R., Bassett, D.S. The network structure of scientific revolutions. *American Physical Society March Meeting* (virtual due to COVID-19). March 15–19, 2021.

[7] **Zhou, D.**, Lynn, C.W., Cui, Z., Ciric, R., Baum, G.L., Moore, T.M., Roalf, D.R., Detre, J.A., Gur, R.C., Gur, R.E., Satterthwaite, T.D., & Bassett, D.S. Efficient Coding in the Economics of Human Brain Connectomics. *Organization for Human Brain Mapping*. Montreal, CA (virtual due to COVID-19). June 23–July 3, 2020.

[6] **Zhou, D.**, Lydon-Staley, D., Zurn, P., Bassett, & D.S. Network Mechanisms of Curiosity and Information Seeking During Wikipedia Exploration. *National Academy of Sciences Colloquium: The Brain Produces the Mind By Modeling*, Beckman Center of the National Academy of Sciences & Engineering, Irvine, California. May 1–3, 2019.

[5] **Zhou, D.**, Liu, S., Zhou, X., Berman, R.A., Broadnax, D.D., Rapoport, J.L, & Thomas, A.G. Ultra-high field 7-Tesla MRI reveals hippocampal subfield volume and shape abnormalities in childhood-onset schizophrenia patients compared to healthy siblings and controls. 9^{th} *Annual Julius Axelrod Symposium*, Bethesda, Maryland. April 13, 2017.

[4] **Zhou, D.**, Liu, S., Zhou, X., Berman, R.A., Broadnax, D.D., Rapoport, J.L, & Thomas, A.G. Ultra-High Field 7-Tesla MRI Shape Analysis of Hippocampal Subfields in Childhood-Onset Schizophrenia and Healthy Siblings. *Society of Biological Psychiatry*, San Diego, California. May 18–20, 2017.

[3] **Zhou, D.**, Liu, S., Berman, R.A., Broadnax, D.D., Rapoport, J.L, & Thomas, A.G. 7-Tesla MRI Reveals Regional Hippocampal Deficits in Childhood-Onset Schizophrenia. *American College of Neuropsychopharmacology*, Hollywood, Florida. In *Neuropsychopharmacology.* December 4–8, 2016.

[2] **Zhou, D.**, Liu, S., Berman, R.A., Broadnax, D.D., Rapoport, J.L, & Thomas, A.G. 7-Tesla MRI reveals regional hippocampal volume deficits of dentate gyrus in childhood-onset schizophrenia. *Society for Neuroscience*, San Diego, California. November 12–16, 2016.

[1] **Zhou, D.**, Gochman, P., Broadnax, D.D., Rapoport, J.L., & Ahn, K. 15q13.3 duplication in two patients with childhood-onset schizophrenia. *Society of Biological Psychiatry*, Atlanta, Georgia. May 12–14, 2016.

TEACHING

Teaching Assistant

2022	Goals of Scientific Inquiry; or, On the Curiosity of Beasts. Instructor: Dani Bassett Student evaluation of TA: 3.6/4.0	*UPenn* Bioengineering
2020	Curiosity: Ancient and Modern Thinking About Thinking. Instructor: Dani Bassett Student evaluation of TA: not rated	*UPenn* Arts & Sciences
2019	Computational Neuroscience Lab. Instructor: Nicole Rust Student evaluation of TA: 3.7/4.0	*UPenn* Psychology

Guest Lecturer

2023–24	Advanced Topics in Graph Analysis (online) Instructor: Will Gray-Roncal	*Johns Hopkins* Engineering; Data Science
2022	Goals of Scientific Inquiry Instructor: Dani Bassett	*UPenn* Bioengineering 571
2019–20	Network Neuroscience Instructor: Dani Bassett	*UPenn* Bioengineering 566
2019	Computational Neuroscience Lab Instructor: Nicole Rust	*UPenn* Biological Basis of Behavior 310

MENTORSHIP

Graduate Students

[3] Gimarie Irizarry Martínez — UC Irvine PhD, Neurobiology & Behavior
Anhedonia, reinforcement learning, value-mediated memory

[2] Bianca Leonard — UC Irvine MD/PhD, Neurobiology & Behavior
Network growth charts, anhedonia, early-life unpredictability; now completing MD

[1] Jungsun Yoo — UC Irvine PhD, Cognitive Science
Cognitive graphs, latent cause models; now Postdoc (Niv & Daw labs, Princeton)

Undergraduate Students

[6] Shuheng (Jerry) Guo — UC Irvine B.S., Psychology
Value-mediated pattern separation, sparse autoencoders, foraging

[5] Destiny Edens — NC A&T State University, Summer Student
Network structure in psychiatric taxonomy; now Teach for America

[4] Emily Dinh — UC Irvine Research Specialist
Cognitive graphs, autoencoders; now Johnson & Johnson

[3] Ivan Tseytlin — Haverford College Physics & Computer Science, 2023
Network control theory; now Daylight Computer Company

[2] Samantha Simon — University of Pennsylvania Physics, 2023
Diversity in science, semantic networks; now Accenture AI

[1] Mark Choi — University of Pennsylvania Computer Science, 2021
Network structure in mathematics; now Meta

SERVICE

Co-organizer & Program Committee

Innovators in Cognitive Neuroscience Symposia | Web Conference (Wiki Workshop) | Organization for Human Brain Mapping Hackathon

Invited Talks

Citation Diversity Tutorial. University of California, Irvine. October 16th, 2023.

Citation Diversity Tutorial, Annual Meeting for the Neuroscience Training Grant; Vision Training Grant; and Computational Approaches to the Neuroscience of Audition Training Grant, University of Pennsylvania. June 7, 2021.

Panelist, Post-Baccalaureate Research Experiences, University of Maryland. March 30, 2017.

Reviewing

Journals: Biological Psychiatry | Cerebral Cortex | Cognitive Science | Communications Biology | Frontiers in Psychiatry | Hippocampus | IEEE Transactions on Network Science and Engineering | Network Neuroscience | npj Artificial Intelligence | PLOS One | Social Cognitive and Affective Neuroscience

Conferences: Association for Computational Linguistics (WikiNLP) | Cognitive Computational Neuroscience | Cognitive Science Society | Flux Society | NeurIPS (Information-Theoretic Principles in Cognitive Systems) | Wiki Workshop

Mentorship

2020–21 MindCORE Step-Ahead Mentorship Program [link]

Mārtiņs M. Gataviņš — University of Pennsylvania, Neuroscience 2024

2019–21 Upward Bound: Research Fridays [link]

OUTREACH

Broader Impacts

2019– Creator/Maintainer, Citation Diversity Statement Code Notebook [link]

- Used and cited in > 150 articles across > 30 journals.
- Highlighted by Nature, Cell, Science, Nature Neuroscience, and Nature Reviews Psychology.

2024 Mentor, Irvine-HBCU Summer Institute in Neuroscience.
2020–22 Organizing Committee, Innovators in Cognitive Neuroscience
2020–21 Web Developer, Black in STEM in Academia
2019–21 Apprentice Chief, Upward Bound: Research Fridays
2019–20 Organizer/Web Developer, Penn Network Visualization Program
2017–20 Section Chief, Brains in Brief Science Communication

Art Exhibitions

[3] Kamen's Lens. Rebecca Kamen, S.J. Fowler, Dale Zhou. The University of Tennessee, Knoxville. 2026. [video 1] [video 2]

[2] Kamen's Lens. Rebecca Kamen, S.J. Fowler, Dale Zhou. *Dyslexic Dictionary*. Organized and curated by Gil Gershoni, Tasmin Smith, and Ted Gioia. Arion Press Gallery, 1802 Hays Street, The Presidio, San Francisco, CA. October 22–December 22, 2022. [link] [video 1] [video 2]

[1] Sparking Curiosity. Dale Zhou, David Lydon-Staley, Perry Zurn, and Dani Bassett. *Reveal: The Art of Reimagining Scientific Discovery*. Organized and curated by Rebecca Kamen and Sarah Tanguy. Museum at the Katzen Arts Center, American University, Washington, D.C., August 29–December 12, 2021. [link]

Media Coverage (Articles & Blogs)

Research featured in university or popular press (authors of articles listed).

[11] Richard Fisher (May 29, 2025) *What style of curiosity do you practise?* Psyche

[10] Gary Stix (Ed. Sarah Frasier) (December 24, 2024) *Wikipedia Searches Reveal Differing Styles of Curiosity. Scientific American*

[9] Nathi Magubane (October 28, 2024) *Studying Wikipedia browsing habits to learn how people learn. Penn Today*

[8] Bronwyn Thompson (October 26, 2024) *What your Wikipedia browsing style says about you. New Atlas*

[7] Sarah Polkinghorne (October 25, 2024) *Going down a Wikipedia rabbit hole? Science says youre one of these three types. The Conversation*

[6] Helena Kudiabor (October 24, 2024) *Study reveals three ways to disappear down a Wikipedia rabbit hole. Nature*

[5] Natalia Gass (February 17, 2023) *Neural states during mindful attention. Nature Mental Health*.

[4] Nathi Magubane (January 26, 2023) *Through the lens: A digital depiction of dyslexia.* *Penn Today*.

[3] Diana Kwon (March 22, 2022) *The rise of citational justice: how scholars are making references fairer.* *Nature*.

[2] Melissa Pappas (January 26, 2021) *Researchers measure different types of curiosity studying hunters and busybodies.* *Penn Today*.

[1] Melissa Pappas (January 12, 2021) *Studying Hunters and Busybodies, Penn and American University Researchers Measure Different Types of Curiosity.* *Penn Engineering Today*.

Media Coverage (Radio & Podcast interviews)

Interviews or featured research (hosts listed).

[3] Lynn Borton (2025). Curiosity and Wikipedia. *Choose to Be Curious*

[2] Dale Drinkwater (November 1, 2024). Wikipedia and Curiosity. *ABC NewsRadio Australia*

[1] Sarah Crespi, Kai Kupferschmidt (October 31, 2024) *The challenges of studying misinformation, and what Wikipedia can tell us about human curiosity.* *Science (podcast)*

REFERENCES

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